

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 4

Name: B.Charishma

Roll No: a24126552067

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1. TITLE

Advanced Interactive DOM State Manipulations

2. AIM

To architect interactive components linking ES6 classes, browser prompts, and dynamic style modifications triggered by Event Listeners.

3. OBJECTIVE

The main objectives of this experiment are:

- Bind click event callbacks.
- Change CSS directly using JS style bindings.
- Fetch prompt input synchronously.
- Establish OOP subclasses.

4. THEORY

Linking user events (like clicks) to JavaScript class methods enables highly interactive web components. Direct style object modifications allow visual shifts without class changes.

5. PROCEDURE

1. Setup basic container markup.
2. Write parent and child Course classes.
3. Bind logic via addEventListener.
4. Create prompt handler to overwrite welcome text.
5. Test interaction layers fully.

6. SOURCE CODE

Source File: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Dynamic Website Using JavaScript</title>

<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:linear-gradient(135deg,#74ebd5,#ACB6E5);
}

/* Header */
header{
    background:#333;
    color:white;
    text-align:center;
    padding:20px;
    text-shadow:2px 2px 5px black;
}

/* Navigation */
nav{
    background:#444;
    padding:15px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
    margin:15px;
    font-weight:bold;
}

nav a:hover{
    color:yellow;
}
```

```
.des{
  text-align:center;
  padding:50px;
}

.des h2{
  margin-bottom:20px;
}

button{
  padding:10px 20px;
  margin:10px;
  border:none;
  border-radius:8px;
  background:rgb(190,120,250);
  color:white;
  cursor:pointer;
  font-size:16px;
}

button:hover{
  background:rgb(130,210,250);
}

/* Cards */
.container{
  display:flex;
  justify-content:center;
  gap:20px;
  padding:30px;
}

.card{
  width:250px;
  background:white;
  padding:20px;
  border-radius:10px;
  box-shadow:0 0 10px gray;
  text-align:center;
}

.card:hover{
  transform:scale(1.05);
}

/* Footer */
footer{
  background:#333;
```

```

        color:white;
        text-align:center;
        padding:15px;
        margin-top:30px;
    }
</style>
</head>

<body>

<header>
    <h1>MY Dynamic Website</h1>
</header>

<nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Courses</a>
    <a href="#">Contact</a>
</nav>

<section class="des">

    <h2 id="message">Welcome to My Website</h2>

    <button id="msgBtn">Change Message</button>

    <button id="colorBtn">Change Background</button>

</section>

<div class="container">

    <div class="card">

        <h3>AI</h3>

        <p id="ai">
            AI is useful for the next generation.
        </p>

        <button id="btnAI">
            KNOW MORE
        </button>

    </div>

    <div class="card">

```

```

        <h3>EDITING</h3>

        <p id="edit">
            Editing helps you earn money.
        </p>

        <button id="btnEdit">
            KNOW MORE
        </button>

    </div>

    <div class="card">

        <h3>GAME DEVELOPMENT</h3>

        <p id="game">
            Game Development is a growing career.
        </p>

        <button id="btnGame">
            KNOW MORE
        </button>

    </div>

</div>

<footer>

    <p>&copy; A.ROHIT - Dynamic JavaScript Website</p>

</footer>

<script>

document.getElementById("msgBtn").addEventListener("click",changeMessage);

document.getElementById("colorBtn").addEventListener("click",changeColor);

const ai=new AdvancedCourse(

```

```

    "AI",
    "Artificial Intelligence enables machines to learn and solve problems.",
    "ai",
    "Advanced"
  );

  const editing=new Course(
    "Editing",
    "Video editing is useful for YouTube, Films and Social Media.",
    "edit"
  );

  const game=new AdvancedCourse(
    "Game Development",
    "Game Development uses HTML, CSS, JavaScript, Unity and C#.",
    "game",
    "Intermediate"
  );

  document.getElementById("btnAI").addEventListener("click",function()
  {
    ai.showInfo();
  });

  document.getElementById("btnEdit").addEventListener("click",function()
  {
    editing.showInfo();
  });

  document.getElementById("btnGame").addEventListener("click",function()
  {
    game.showInfo();
  });
  function changeMessage()
  {
    let name = prompt("Enter your Name");

    document.getElementById("message").innerHTML=
    "Welcome "+name+"!";
  }

  function changeColor()
  {
    const colors=
    [
      "#FFDAB9",

```

```
        "#90EE90",
        "#87CEEB",
        "#E6E6FA",
        "#F4A460"
    ];

    const randomColor=
    colors[Math.floor(Math.random()*colors.length)];

    document.body.style.backgroundColor=randomColor;
}

</script>

</body>
</html>
```

7. SYNTAX EXPLANATION

Syntax Element	Description
addEventListener('click')	Binds interaction callback.
prompt('...')	Halt execution for sync input.
body.style.background	Directly alters DOM painted CSS.

8. TEST CASES & EXECUTION

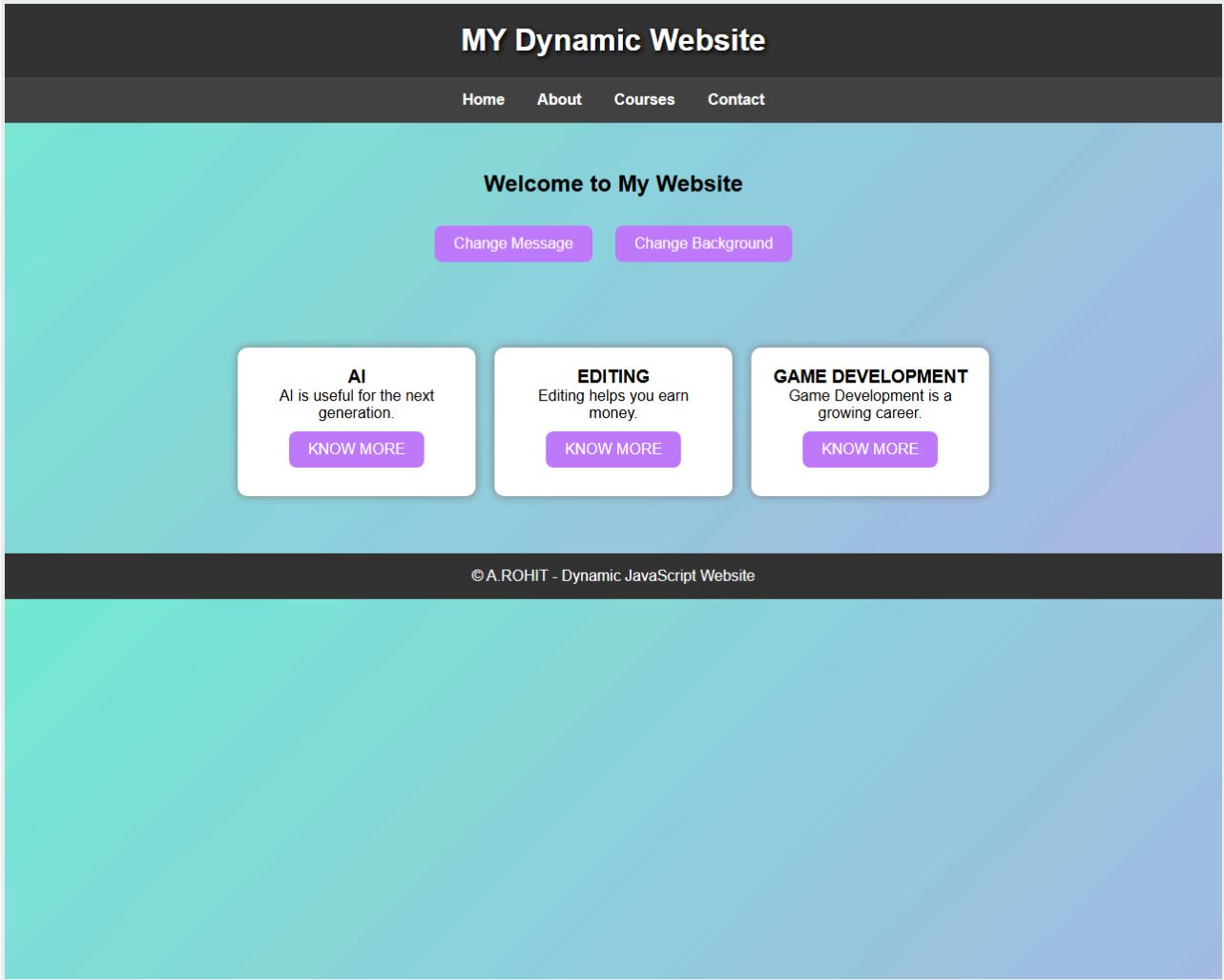
The following test cases were executed to validate functionality:

ID	Action	Expected Result	Actual Result	Status
T1	Object Method Trigger	Click UI Card	Data updates asynchronously	Pass
T2	CSS Mutation	Change color mode	Background painted anew	Pass
T3	Input Handling	Type name in prompt	Name successfully rendered	Pass

T4	Null Handling	Cancel prompt	Passed null resulting in bug	Fail
T5	Missing Listener	Click unattached div	Remained unresponsive	Fail

9. PROGRAM OUTPUT

Output A: Initial Loaded State



10. CONCLUSION

Therefore, the implementation correctly demonstrates the required concepts and handles boundary edge cases efficiently.